

QUESTION ONE (6 Marks)**MULTIPLE CHOICE:** Write the correct alternative on your writing paper.

1. Which of the following is an expression for $\int x e^{2x} dx$? **1**
 - (A) $e^{2x} \left(\frac{x}{2} - \frac{1}{4} \right) + c$
 - (B) $e^{2x} \left(\frac{x}{2} - 1 \right) + c$
 - (C) $e^{2x} \left(x - \frac{1}{4} \right) + c$
 - (D) $e^{2x} (2x - 1) + c$

2. Which of the following is the primitive function of $\frac{\cos \sqrt{x}}{\sqrt{x}}$? **1**
 - (A) $\sin \sqrt{x} + c$
 - (B) $-\sin \sqrt{x} + c$
 - (C) $\frac{1}{2} \cos^2 \sqrt{x} + c$
 - (D) $2 \sin \sqrt{x} + c$

3. Which of the following is the primitive function of $\frac{1}{\sqrt{4-9x^2}}$? **1**
 - (A) $\frac{1}{6} \sin^{-1} \frac{3x}{2} + c$
 - (B) $\frac{1}{3} \sin^{-1} \frac{3x}{2} + c$
 - (C) $\frac{2}{3} \sin^{-1} \frac{3x}{2} + c$
 - (D) $\frac{1}{3} \sin^{-1} \frac{2x}{3} + c$

4. Which of the following is equal to $\int_{-2}^2 \sqrt{4-x^2} \, dx$? 1

(A) 0

(B) π

(C) 2π

(D) 4π

5. Which of the following is the primitive function of $\int \frac{1-\ln x}{x^2} \, dx$? 1

(A) $\frac{\ln x}{x} + c$

(B) $x \ln x + c$

(C) $\ln x^2 + c$

(D) $\frac{x}{\ln x} + c$

6. Which of the following statements are true ? 1

1. $\int_{-\pi}^{\pi} \sin^7 \theta \, d\theta = 0$

2. $\int_{-\pi}^{\pi} \cos^7 \theta \, d\theta = 0$

(A) Both are true.

(B) Both are false.

(C) Only 1 is true.

(D) Only 2 is true.

QUESTION TWO (26Marks) Start a new page

(a) Evaluate $\int_0^1 \frac{dx}{\sqrt{4x^2 + 36}}$. 2

(b) Find $\int \frac{\cos \theta}{\sin^5 \theta} d\theta$. 1

(c) Using the substitution $t = \tan \frac{x}{2}$, evaluate $\int_0^{\frac{\pi}{2}} \frac{d\theta}{3 \sin \theta + 4 \cos \theta}$. 3

(d) Find $\int \frac{1}{x^2 - 12x + 61} dx$. 2

(e) Use the substitution $x = 3 \sin \theta$, to evaluate $\int_0^{\frac{3}{\sqrt{2}}} \frac{dx}{(9 - x^2)^{\frac{3}{2}}}$. 4

(f) (i) Find the real numbers a and b such that 2

$$\frac{x^2 - 7x + 4}{(x+1)(x-1)^2} = \frac{a}{x+1} + \frac{b}{x-1} - \frac{1}{(x-1)^2}.$$

(ii) Hence find $\int \frac{x^2 - 7x + 4}{(x+1)(x-1)^2} dx$. 2

(g) Find $\int \sqrt{\frac{x+1}{x-1}} dx$. 2

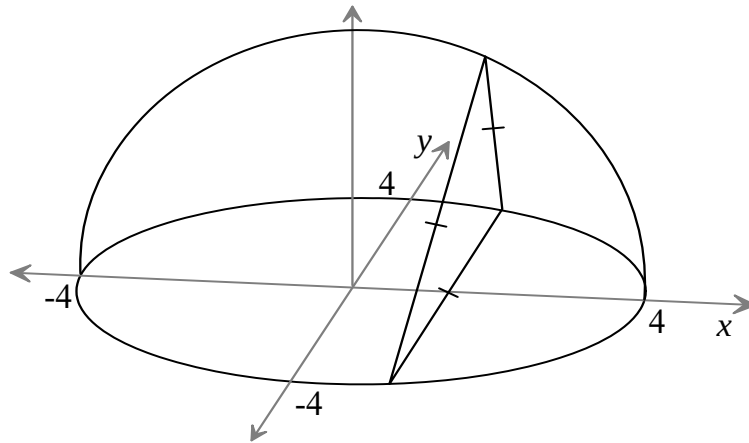
(h) (i) Let $I_n = \int (\ln x)^n dx$, show that $I_n = x(\ln x)^n - nI_{n-1}$. 5

(ii) Hence evaluate $\int_1^{e^4} (\ln x)^3 dx$.

(i) Find $\int 2 \sin \theta \cos \theta (3 \sin \theta - 4 \sin^3 \theta) d\theta$ 3

QUESTION THREE (12 Marks) Start a new page

- (a) The region enclosed by the curve $y = 5x - x^2$, the x axis and the lines $x = 1$ and $x = 3$ is rotated about the y axis. By using the method of cylindrical shells, find the volume of the solid so produced. 4
- (b) (i) Sketch the region bounded by the curve $y = \log_e x$, the x -axis and the vertical line $x = e$. 4
- (ii) The region is rotated about the y -axis to form a solid. Find the volume of the solid by slicing perpendicular to the axis of rotation.
- (c) 4



The diagram above shows a solid which has the circle $x^2 + y^2 = 16$ as its base. The cross-section perpendicular to the x axis is an equilateral triangle. Calculate the volume of the solid.

END OF PAPER

Question 1

$$1. |z| = 2\sqrt{5} \quad \arg z = -\pi + \tan^{-1} 2$$

z lies in the 3rd quadrant

$$\therefore z = -2 - 4i \quad (C)$$

$$2. \text{ Let } z^3 = -8$$

$$z^3 + 8 = 0$$

$$(z+2)(z^2-2z+4) = 0$$

$$(z+2)[(z-1)^2+3] = 0$$

$$(z+2)(z-1+\sqrt{3}i)(z-1-\sqrt{3}i) = 0$$

$$\therefore z = -2, 1-\sqrt{3}i, 1+\sqrt{3}i \quad (A)$$

$$3. 3x^2 + (2-i)x + (4+i) = 0$$

$$\alpha + \beta = -\frac{b}{a}$$

$$= -\frac{2+i}{3}$$

$$\alpha\beta = \frac{c}{a}$$

$$= \frac{4+i}{3}$$

$$\frac{1}{z} + \frac{1}{w} = \frac{w+z}{wz}$$

$$= \frac{-\frac{2+i}{3}}{\frac{4+i}{3}}$$

$$= \frac{-2+i}{4+i}$$

$$= \frac{-2+i}{4+i} \times \frac{4-i}{4-i}$$

$$= \frac{-8+6i-i^2}{17}$$

$$= \frac{-7+6i}{17}$$

$$\therefore \frac{1}{z} + \frac{1}{w} = \frac{-7-6i}{17} \quad (A)$$

Question 2

$$a) z = 3+i \quad w = 2-5i$$

$$\begin{aligned} \text{i) } z^2 &= (3+i)^2 \\ &= 9+6i+i^2 \\ &= 8+6i \end{aligned}$$

$$\begin{aligned} \text{ii) } \bar{z} + w &= 3-i+2-5i \\ &= 5-6i \end{aligned}$$

$$\begin{aligned} \text{iii) } \frac{w}{z} &= \frac{2-5i}{3+i} \\ &= \frac{2-5i}{3+i} \times \frac{3-i}{3-i} \\ &= \frac{6-17i+5i^2}{10} \\ &= \frac{1}{10} - \frac{17}{10}i \end{aligned}$$

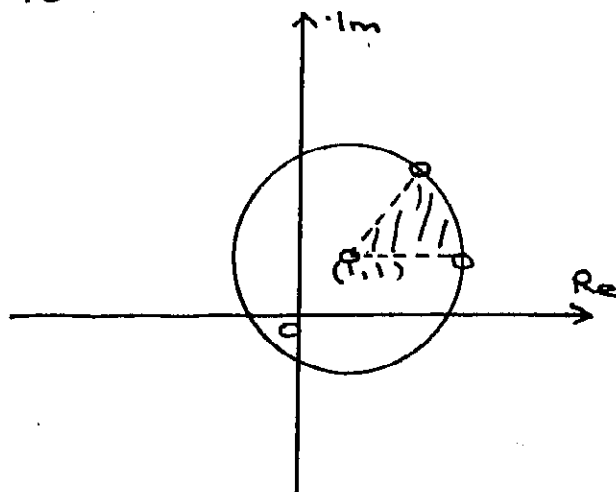
$$b) (x+iy)^2 = 24+10i$$

$$x^2 - y^2 = 24$$

$$xy = 5$$

$\therefore$ The 2 square roots are $\pm(5+i)$

$$c) |z-1-i| \leq 2 \quad 0 < \arg(z-1-i) < \frac{\pi}{4}$$



d) u) $z = 1+i$

$$|z| = \sqrt{1^2 + 1^2} \\ = \sqrt{2}$$

$$\arg z = \tan^{-1} 1 \quad -\pi < \theta \leq \pi \\ = \frac{\pi}{4}$$

$$\therefore 1+i = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$$

$$(1+i)^{17} = (\sqrt{2})^{17} \left(\cos \frac{17\pi}{4} + i \sin \frac{17\pi}{4} \right)$$

$$= 256\sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$$

$$= 256 + 256i$$

e) Show $(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta) \quad n \geq 1$

Step 1

Test $n=1$

$$\text{L.H.S} = (\cos \theta + i \sin \theta)^1$$

$$= \cos \theta + i \sin \theta$$

$$\text{R.H.S} = \cos \theta + i \sin \theta$$

$$\text{L.H.S} = \text{R.H.S}$$

$\therefore$ Result is true for $n=1$

Step 2

Assume that the result is true for $n=k$
that is $(\cos \theta + i \sin \theta)^k = \cos(k\theta) + i \sin(k\theta)$

Step 3

Hence show the result is true for $n=k+1$
that is $(\cos \theta + i \sin \theta)^{k+1} = \cos(k+1)\theta + i \sin(k+1)\theta$

$$\text{L.H.S} = (\cos \theta + i \sin \theta)^{k+1}$$

$$= (\cos \theta + i \sin \theta)^k (\cos \theta + i \sin \theta)$$

$$= (\cos k\theta + i \sin k\theta) (\cos \theta + i \sin \theta) \text{ by step 2}$$

$$= \cos k\theta \cos \theta + i \sin \theta \cos k\theta + i \sin k\theta \cos \theta \\ + i^2 \sin \theta \sin k\theta$$

$$= (\cos k\theta \cos \theta - \sin k\theta \sin \theta) +$$

$$i(\cos k\theta \sin \theta + \sin k\theta \cos \theta)$$

$$= \cos(k+1)\theta + i \sin(k+1)\theta$$

$$= \text{R.H.S}$$

Step 4

Since the result is true for $n=1$ then

from Step 3 it is true for $n=k+1=2$,
and then for $n=3$, and so on by the
process of mathematical induction it is true for
all positive integral values of n

$$e) \quad \omega^3 = 1$$

$$\omega^3 - 1 = 0$$

$$(\omega - 1)(\omega^2 + \omega + 1) = 0$$

$$\omega - 1 \neq 0 \quad \therefore \omega^2 + \omega + 1 = 0$$

$$(i) \quad (1 + 2\omega + 3\omega^2)(1 + 3\omega + 2\omega^2)$$

$$= 1 + 3\omega + 2\omega^2 + 2\omega + 6\omega^2 + 4\omega^3 + 3\omega^2 + 9\omega^3 + 6\omega^4$$

$$= 1 + 5\omega + 11\omega^2 + 13\omega^3 + 6\omega^4$$

$$= 1 + 5\omega + 11\omega^2 + 13 + 6\omega$$

$$\text{NB } \textcircled{1} \omega^3 = 1$$

$$= 14 + 11\omega + 11\omega^2$$

$$\textcircled{2} \omega^4 = \omega$$

$$= 14 + 11(\omega^2 + \omega)$$

$$\textcircled{3} \omega^2 + \omega + 1 = 0$$

$$= 14 - 11$$

$$\omega^2 + \omega = -1$$

$$= 3$$

$$(ii) \quad 1 + 2\omega + 3\omega^2 + 1 + 3\omega + 2\omega^2 = 2 + 5\omega + 5\omega^2$$

$$= 2 + 5(\omega + \omega^2)$$

$$= 2 - 5$$

$$= -3$$

(iii) From parts (i) and (ii) $(1 + 2\omega + 3\omega^2)$ and $(1 + 3\omega + 2\omega^2)$ are roots of $x^2 + 3x + 3 = 0$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-3 \pm \sqrt{3^2 - 4 \times 1 \times 3}}{2 \times 1}$$

$$= \frac{-3 \pm \sqrt{3}}{2}$$

$$\therefore 1 + 2\omega + 3\omega^2 = \frac{-3 - \sqrt{3}}{2}$$

$$\text{NB } \text{Im}(\omega) = -\text{Im}(\omega^2)$$

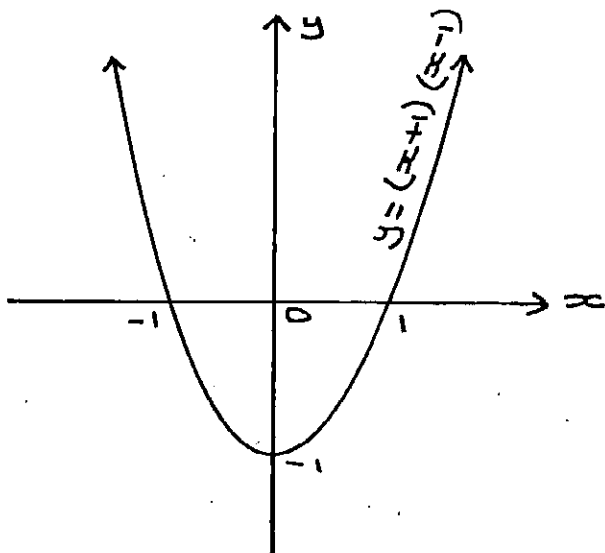
$$1 + 3\omega + 2\omega^2 = \frac{-3 + \sqrt{3}}{2}$$

$$\text{Im}(2\omega + 3\omega^2) = \text{Im}(\omega^2) < 0$$

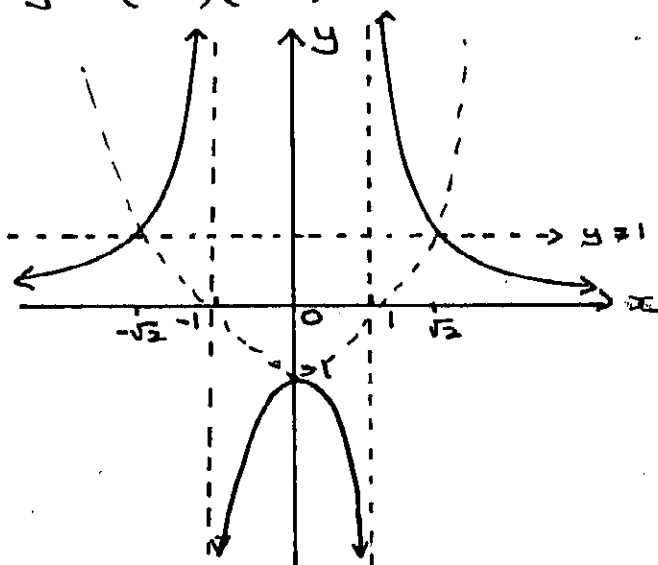
$$\text{Im}(3\omega + 2\omega^2) = \text{Im}(\omega) > 0$$

Question 3

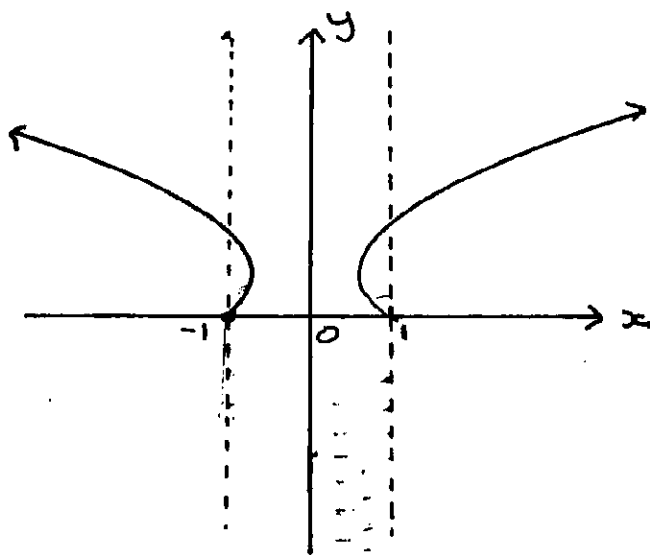
a) (i) $y = (x+1)(x-1)$



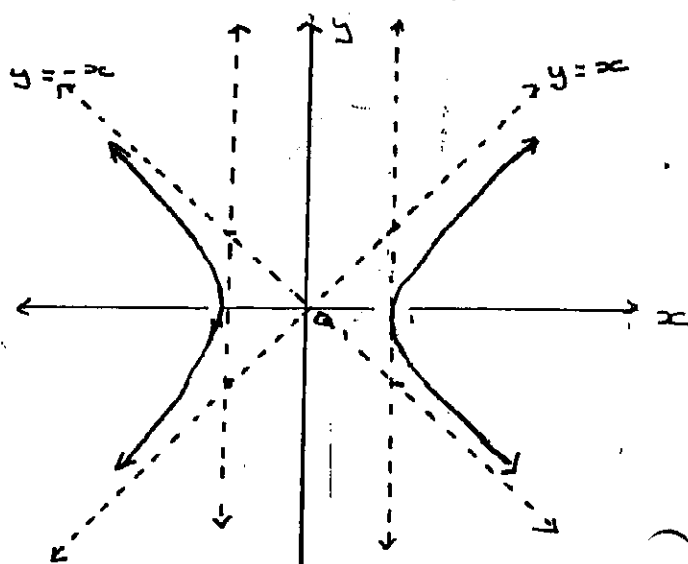
ii) $y = \frac{1}{(x-1)(x+1)}$



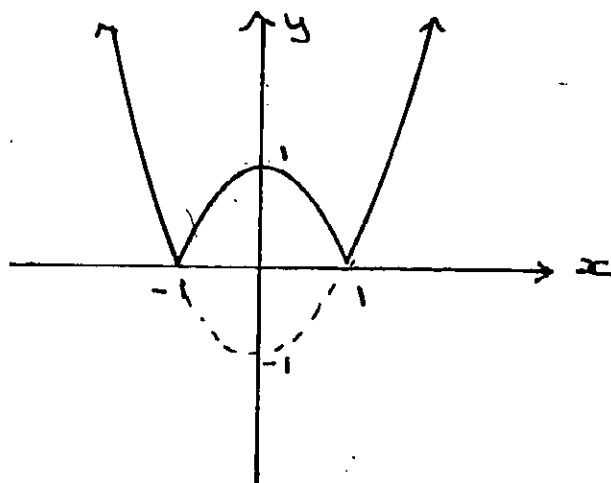
iii) $y = \sqrt{(x-1)(x+1)}$



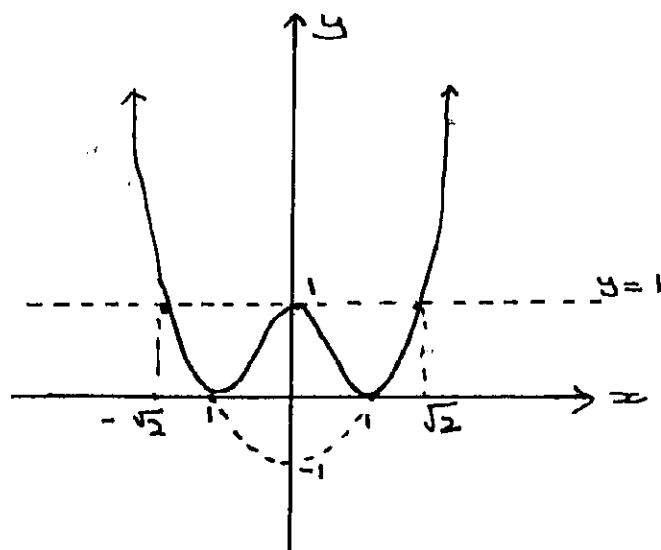
(v) $y^2 = (x+1)(x-1)$



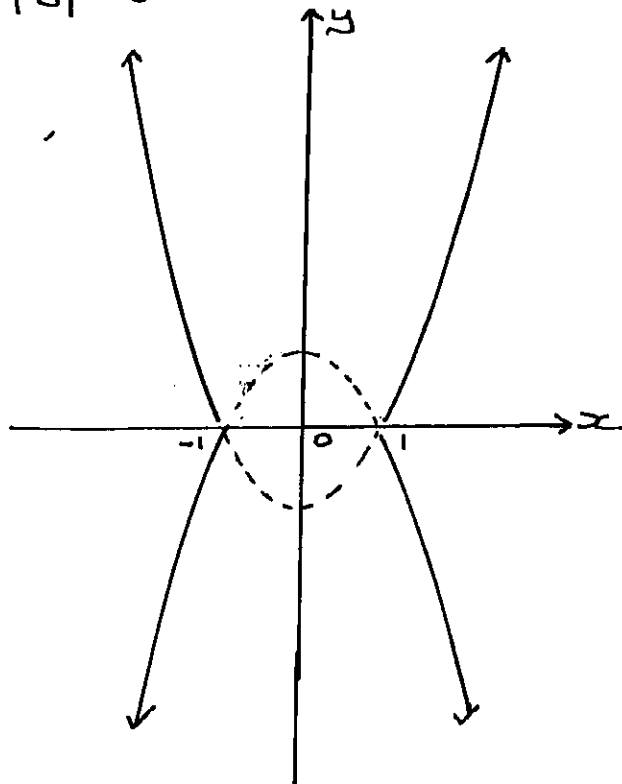
vi) $y = |(x-1)(x+1)|$



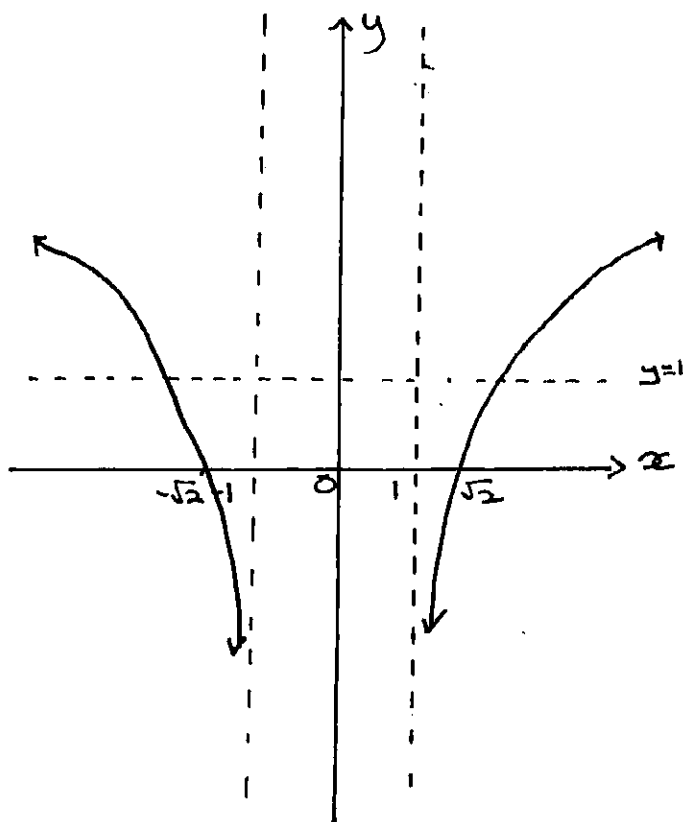
(vi) $y = [(x+1)(x-1)]^2$



vii) $|y| = (x-1)(x+1)$



viii) $y = \log_e (x+1)(x-1)$



b) $f(x) = \frac{x^4}{x^2-1}$

$$f(-x) = \frac{(-x)^4}{(-x)^2-1}$$

$$= \frac{x^4}{x^2-1}$$

$$\therefore f(x) = f(-x)$$

$\therefore f(x) = \frac{x^4}{x^2-1}$ is an even function.

ii) $f(x) = \frac{x^4}{x^2-1}$

$$u = x^4 \quad v = x^2-1$$

$$\frac{du}{dx} = 4x^3 \quad \frac{dv}{dx} = 2x$$

$$f'(x) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$= \frac{(x^2-1)4x^3 - x^4 \times 2x}{(x^2-1)^2}$$

$$= \frac{2x^3(2x^2-2-x^2)}{(x^2-1)^2}$$

$$= \frac{2x^3(x^2-2)}{(x^2-1)^2}$$

$$= \frac{2x^3(x-\sqrt{2})(x+\sqrt{2})}{(x^2-1)^2}$$

For stationary points

$$f'(x) = 0$$

$$\therefore 2x^3(x-\sqrt{2})(x+\sqrt{2}) = 0$$

$$\therefore x = 0, \pm\sqrt{2}$$

$$f(x) = \frac{x^4}{x^2-1}$$

$$f(0) = 0$$

$$f(\sqrt{2}) = \frac{(\sqrt{2})^4}{(\sqrt{2})^2-1}$$

$$= 4$$

$$f(-\sqrt{2}) = \frac{(-\sqrt{2})^4}{(-\sqrt{2})^2-1}$$

$$= 4$$

$\therefore$ The stationary points are $(0,0)$ $(\sqrt{2}, 4)$ and $(-\sqrt{2}, 4)$

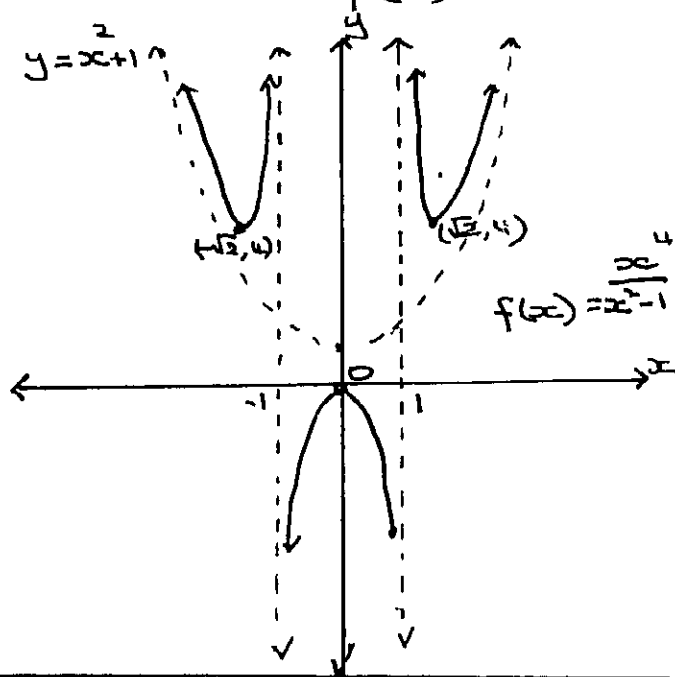
(iii) vertical asymptotes are $x = \pm 1$

$$f(x) = \frac{x^4}{x^2-1}$$

$$= x^2 + 1 + \frac{1}{x^2-1}$$

$$\text{as } x \rightarrow \infty \quad f(x) \rightarrow x^2 + 1$$

$$\text{as } x \rightarrow -\infty \quad f(x) \rightarrow x^2 + 1$$



$$c) \quad 3x^2 + y^2 - 2xy - 8x + 2 = 0 \quad \text{--- ①}$$

$$6x + 2y \frac{dy}{dx} - (2x \frac{dy}{dx} + 2y) - 8 = 0$$

$$(2y - 2x) \frac{dy}{dx} = 8 + 2y - 6x$$

$$\frac{dy}{dx} = \frac{8 + 2y - 6x}{2y - 2x}$$

$$\therefore \frac{dy}{dx} = \frac{4 + y - 3x}{y - x}$$

(iii) As the tangent is parallel to $y = 2x$, $\frac{dy}{dx} = 2$

$$\frac{4 + y - 3x}{y - x} = 2$$

$$4 + y - 3x = 2y - 2x$$

$$y = 4 - x \quad \text{--- ②}$$

sub ② in ①

$$3x^2 + y^2 - 2xy - 8x + 2 = 0$$

$$3x^2 + (4-x)^2 - 2x(4-x) - 8x + 2 = 0$$

$$3x^2 + 16 - 8x + x^2 - 8x + 2x^2 - 8x + 2 = 0$$

$$6x^2 - 24x + 18 = 0$$

$$x^2 - 4x + 3 = 0$$

$$(x-3)(x-1) = 0$$

$$x = 3, 1$$

when $x = 3$

$$\text{sub ② } y = 4 - x$$

$$= 4 - 3$$

$$= 1$$

when $x = 1$

$$\text{sub ② } y = 4 - x$$

$$= 4 - 1$$

$$= 3$$

$\therefore$ The points are $(3, 1)$ $(1, 3)$

Question 4

$$\begin{aligned}
 1. \quad P(x) &= x^4 - 1 \\
 &= (x^2 - 1)(x^2 + 1) \\
 &= (x-1)(x+1)(x-1)(x+1) \\
 &\quad (C)
 \end{aligned}$$

$$2. \quad x^3 - 8x^2 - 4x + 12 = 0$$

$$\begin{aligned}
 \alpha + \beta + \gamma &= \frac{-b}{a} \\
 &= 8
 \end{aligned}$$

$$\begin{aligned}
 \alpha\beta + \alpha\gamma + \beta\gamma &= \frac{c}{a} \\
 &= -4
 \end{aligned}$$

$$\begin{aligned}
 \alpha\beta\gamma &= \frac{-d}{a} \\
 &= -12
 \end{aligned}$$

$$\begin{aligned}
 \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} &= \frac{\alpha\beta + \alpha\gamma + \beta\gamma}{\alpha\beta\gamma} \\
 &= \frac{-4}{-12} \\
 &= \frac{1}{3} \quad (C)
 \end{aligned}$$

$$3. \quad \frac{1}{(x+1)(x-3)} = \frac{A}{x+1} + \frac{B}{x-3}$$

$$1 = A(x-3) + B(x+1)$$

$$\text{Sub } x=3$$

$$1 = B(3+1)$$

$$B = \frac{1}{4}$$

$$\text{Sub } x=-1$$

$$1 = A(-1-3)$$

$$A = -\frac{1}{4}$$

$$\therefore A = -\frac{1}{4}, B = \frac{1}{4} \quad (D)$$

Question 5

$$a) \quad x^3 - 4x^2 + 5x + 2 = 0$$

$$\begin{aligned}
 \alpha + \beta + \gamma &= \frac{-b}{a} \\
 &= 4
 \end{aligned}$$

$$\begin{aligned}
 \alpha\beta + \alpha\gamma + \beta\gamma &= \frac{c}{a} \\
 &= 5
 \end{aligned}$$

$$\begin{aligned}
 \alpha\beta\gamma &= \frac{-d}{a} \\
 &= -2
 \end{aligned}$$

$$\begin{aligned}
 a) \quad \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \\
 &= 4^2 - 2 \times 5 \\
 &= 6
 \end{aligned}$$

$$(ii) \quad \alpha^3 = 4\alpha^2 - 5\alpha - 2 \quad \text{--- (1)}$$

$$\beta^3 = 4\beta^2 - 5\beta - 2 \quad \text{--- (2)}$$

$$\gamma^3 = 4\gamma^2 - 5\gamma - 2 \quad \text{--- (3)}$$

$$\textcircled{1} + \textcircled{2} + \textcircled{3}$$

$$\begin{aligned}
 \alpha^3 + \beta^3 + \gamma^3 &= 4(\alpha^2 + \beta^2 + \gamma^2) - 5(\alpha + \beta + \gamma) - 2 \times 3 \\
 &= 4(6) - 5(4) - 6 \\
 &= -2
 \end{aligned}$$

$$b) \quad 3x^4 + 17x^3 + 30x^2 + 12x - 8 = 0$$

$$P(x) = 3x^4 + 17x^3 + 30x^2 + 12x - 8$$

$$P'(x) = 12x^3 + 51x^2 + 60x + 12$$

$$P''(x) = 36x^2 + 102x + 60$$

$$= 2(18x^2 + 51x + 30)$$

$$= 2(x+2)(18x+15)$$

$$P(-2) = 0$$

$$P'(-2) = 0$$

$$P''(-2) = 0$$

$$\therefore (x+2) \text{ is a triple root}$$

$$\therefore 3x^4 + 17x^3 + 30x^2 + 12x - 8 = 0$$

$$(x+2)^3(3x-1) = 0$$

by inspection

c) By De Moivre's Theorem $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$

i) $(\cos \theta + i \sin \theta)^3 = \cos 3\theta + i \sin 3\theta$

$$\begin{aligned} \text{L.H.S} &= (\cos \theta + i \sin \theta)^3 \\ &= \cos^3 \theta + 3\cos^2 \theta i \sin \theta + 3\cos \theta (i \sin \theta)^2 + i^3 \sin^3 \theta \\ &= \cos^3 \theta + 3\cos^2 \theta i \sin \theta - 3\cos \theta \sin^2 \theta - i \sin^3 \theta \\ &= \cos^3 \theta - 3\cos \theta \sin^2 \theta + i(3\cos^2 \theta \sin \theta - \sin^3 \theta) \end{aligned}$$

Equating real parts

$$\begin{aligned} \cos 3\theta &= \cos^3 \theta - 3\cos \theta \sin^2 \theta \\ &= \cos^3 \theta - 3\cos \theta (1 - \cos^2 \theta) \\ &= \cos^3 \theta - 3\cos \theta + 3\cos^3 \theta \end{aligned}$$

$$\therefore \cos 3\theta = 4\cos^3 \theta - 3\cos \theta$$

ii) $8x^3 - 6x - 1 = 0$

$$4x^3 - 3x = \frac{1}{2}$$

But $\cos 3\theta = \frac{1}{2}$

NB $3\theta = \pm \frac{\pi}{3} + 2n\pi$
 n is integral

$$3\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}$$

$$\theta = \frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}$$

$\therefore$ The roots of $8x^3 - 6x - 1 = 0$ are $\cos \frac{\pi}{9}$,

$$\cos \frac{5\pi}{9} = -\cos \frac{4\pi}{9} \text{ and } \cos \frac{7\pi}{9} = -\cos \frac{2\pi}{9}$$

iii) $8x^3 - 6x - 1 = 0$

$$\begin{aligned} \text{product of the roots} &= \frac{-d}{a} \\ &= \frac{1}{8} \end{aligned}$$

$$\therefore \cos \frac{\pi}{9} \times -\cos \frac{4\pi}{9} \times -\cos \frac{2\pi}{9} = \frac{1}{8}$$

$$\therefore \cos \frac{\pi}{9} \cos \frac{2\pi}{9} \cos \frac{4\pi}{9} = \frac{1}{8}$$

$$d) \frac{2x+31}{(x-1)^3(x+2)} = \frac{A}{(x-1)} + \frac{B}{(x-1)^2} + \frac{C}{(x-1)^3} + \frac{D}{(x+2)}$$

$$2x+31 = A(x-1)^2(x+2) + B(x-1)(x+2) + C(x+2) + D(x-1)^3$$

Let $x=1$ $2(1)+31 = C(1+2)$

$$33 = 3C$$

$$C = 11$$

Let $x=-2$ $2(-2)+31 = D(-2-1)^3$

$$27 = -27D$$

$$D = -1$$

Let $x=0$ $2(0)+31 = A(-1)^2(2) + B(-1)(2) + 11(2) - 1(-1)^3$

$$31 = 2A - 2B + 22 + 1$$

$$8 = 2A - 2B$$

$$4 = A - B \quad \text{--- (1)}$$

Let $x=-1$ $2(-1)+31 = A(-1-1)^2 + B(-1-1)(-1+2) + 11(-1+2) - 1(-1-1)^3$

$$29 = 4A - 2B + 11 + 8$$

$$10 = 4A - 2B$$

$$5 = 2A - B \quad \text{--- (2)}$$

Solve (1) and (2) simultaneously

$$A - B = 4 \quad \text{--- (1)}$$

$$2A - B = 5 \quad \text{--- (2)}$$

$$\textcircled{1} - \textcircled{2}$$

$$-A = -1$$

$$A = 1$$

Sub (1)

$$A - B = 4$$

$$1 - B = 4$$

$$B = -3$$

$$\therefore A = 1, B = -3, C = 11, D = -1$$

$$e) \quad x^3 + 3px + q = 0$$

$$\alpha + \beta + \gamma = \frac{-b}{a} \\ = 0$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} \\ = 3p$$

$$\alpha\beta\gamma = \frac{-d}{a} \\ = -q$$

$$\begin{aligned} \text{sum of the roots} &= \frac{\alpha\beta}{\gamma} + \frac{\beta\gamma}{\alpha} + \frac{\alpha\gamma}{\beta} \\ &= \frac{\alpha^2\beta^2 + \beta^2\gamma^2 + \alpha^2\gamma^2}{\alpha\beta\gamma} \\ &= \frac{(\alpha\beta + \beta\gamma + \alpha\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)}{\alpha\beta\gamma} \\ &= \frac{(3p)^2 - q(0)}{-q} \\ &= \frac{9p^2}{-q} \end{aligned}$$

$$\begin{aligned} \text{sum of the roots 2 at a time} &= \frac{\alpha\beta}{\gamma} \times \frac{\beta\gamma}{\alpha} + \frac{\beta\gamma}{\alpha} \times \frac{\gamma\alpha}{\beta} + \frac{\gamma\alpha}{\beta} \times \frac{\alpha\beta}{\gamma} \\ &= \beta^2 + \gamma^2 + \alpha^2 \\ &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \\ &= 0 - 2(3p) \\ &= -6p \end{aligned}$$

$$\begin{aligned} \text{Product of the roots} &= \frac{\alpha\beta}{\gamma} \times \frac{\alpha\gamma}{\beta} \times \frac{\beta\gamma}{\alpha} \\ &= -q \end{aligned}$$

$$\therefore \text{The monic equation is } x^3 + \frac{9p^2}{q}x - 6px + q = 0$$

(ii) If $\gamma = \alpha\beta$ then one of the roots, $\frac{\alpha\beta}{\gamma} = \frac{\gamma}{\gamma} = 1$

$$\therefore x^3 + \frac{9p^2}{q}x^2 - 6px + q = 0$$

$$1 + \frac{9p^2}{q} - 6p + q = 0$$

$$q + 9p^2 - 6pq + q^2 = 0$$

$$q + (3p - q)(3p - q) = 0$$

$$\therefore (3p - q)^2 + q = 0$$